

Analysis 2: HW1

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Problem 1.

1. We are given $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ which is Frechet differentiable everywhere and $f' = 0$. First write f as it's component functions $f = (f_1, \dots, f_m)$. Now for each component function f_i let us apply MVT. Consider $a, b \in \mathbb{R}^n$, then as each f_i is a real valued function if we restrict f_i to the line segment joining a, b then we can apply MVT to f_i defined on that parametrized line i.e $a + (b - a)t$ to get,

$$f_i(b) - f_i(a) = \Delta f_i(c)(b - a)$$

where c is some point on the line segment connecting a, b . However, we are given that $\Delta f_i(x) = 0$ for all points x in the domain, hence we have $\Delta f_i(c) = 0$ and equivalently,

$$f_i(b) = f_i(a)$$

So for each component function we have shown that that function must be constant throughout, and hence our function $f = (f_1, \dots, f_m)$ must also be constant throughout.

2. We are now given that f' is a constant matrix A , using the same formulation as above we have,

$$f_i(b) - f_i(a) = \Delta f_i(c)(b - a)$$

However, now we have $\Delta f_i(c)$ is a constant vector for any c . This implies that our component function f_i is an affine function. Hence, $f = (f_1, \dots, f_m)$ is a affine transformation as well. i.e we have $f(x) = Ax + b$ for some b and where $f'(x) = A$. □

Problem 2 We have $\lim_{r \downarrow 0} r^\alpha \ln \left(10 + \frac{1}{r}\right)^\beta$. Rewriting it we get,

$$\begin{aligned} \lim_{r \downarrow 0} r^\alpha \ln \left(10 + \frac{1}{r}\right)^\beta &= \lim_{r \downarrow 0} r^\alpha \beta \ln \left(10 + \frac{1}{r}\right) \\ &= \lim_{r \downarrow 0} \frac{\beta \ln \left(10 + \frac{1}{r}\right)}{\frac{1}{r^\alpha}} \end{aligned}$$

Now note that this is of the form $\frac{f(x)}{g(x)}$ where $f(x) \rightarrow \infty$ and $g(x) \rightarrow \infty$. So we

can use L'Hospital's rule to get,

$$\begin{aligned}\lim_{r \downarrow 0} \frac{\beta \ln(10 + \frac{1}{r})}{\frac{1}{r^\alpha}} &= \lim_{r \downarrow 0} \frac{\beta \frac{1}{10 + \frac{1}{r}} \frac{-1}{r^2}}{-\alpha \frac{1}{r^{\alpha+1}}} \\ &= \lim_{r \downarrow 0} \frac{\beta}{\alpha} \frac{r^{\alpha+1}}{10r^2 + r}\end{aligned}$$

Now note this is of the form $f(x)/g(x)$ again where $f(x) \rightarrow 0$ and $g(x) \rightarrow 0$ so this is equal to,

$$\lim_{r \downarrow 0} \frac{\beta}{\alpha} \frac{r^{\alpha+1}}{10r^2 + r} = \lim_{r \downarrow 0} \frac{\beta}{\alpha} \frac{(\alpha+1)r^\alpha}{20r + 1}$$

Clearly as $r \rightarrow 0^+$ we have the numerator goes to 0 (as $\alpha > 0$ we have $r^\alpha \rightarrow 0$ when $r \rightarrow 0$) and the denominator goes to 1 and hence we have the limit is equal to 0.

Problem 3. We are given $f : \mathbb{R} \rightarrow \mathbb{R}$ with f'' as a continuous function and f is non-constant. To show the conditions are equivalent it is enough to show that (1) $\Rightarrow$ (3), (3) $\Rightarrow$ (2) and (2) $\Rightarrow$ (3) as all the other equivalencies then follow.

(1) $\Rightarrow$ (3) We are given $f'' \geq 0$. Now Taylors theorem in the second order states that,

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2}f''(c) \quad c \text{ between } a \text{ and } a+h$$

First we take $a = \alpha x + (1-\alpha)y$ and $h = x - (\alpha x + (1-\alpha)y) = x(1-\alpha) - (1-\alpha)y = (1-\alpha)(x-y)$ and we have,

$$f(x) = f(\alpha x + (1-\alpha)y) + (1-\alpha)(x-y)f'(\alpha x + (1-\alpha)y) + \frac{h^2}{2}f''(c_1)$$

Similarly we take $a = \alpha x + (1-\alpha)y$ and $h = y - (\alpha x + (1-\alpha)y) = y - (1-\alpha)y - \alpha x = \alpha(y-x)$

$$f(y) = f(\alpha x + (1-\alpha)y) + \alpha(y-x)f'(\alpha x + (1-\alpha)y) + \frac{h^2}{2}f''(c_2)$$

Now as $f'' > 0$ we can write this as,

$$f(x) \geq f(\alpha x + (1-\alpha)y) + (1-\alpha)(x-y)f'(\alpha x + (1-\alpha)y) \quad (1)$$

$$f(y) \geq f(\alpha x + (1-\alpha)y) + \alpha(y-x)f'(\alpha x + (1-\alpha)y) \quad (2)$$

Multiplying (1) with α and (2) with $(1-\alpha)$ we get,

$$\begin{aligned}\alpha f(x) &\geq \alpha f(\alpha x + (1-\alpha)y) + \alpha(1-\alpha)(x-y)f'(\alpha x + (1-\alpha)y) \\ (1-\alpha)f(y) &\geq (1-\alpha)f(\alpha x + (1-\alpha)y) + (1-\alpha)\alpha(x-y)f'(\alpha x + (1-\alpha)y)\end{aligned}$$

Summing the two we have,

$$\begin{aligned}\alpha f(x) + (1-\alpha)f(y) &\geq (\alpha + 1-\alpha)f(\alpha x + (1-\alpha)y) + \alpha(1-\alpha)((x-y) + (x-y))f'(\alpha x + (1-\alpha)y) \\ \alpha f(x) + (1-\alpha)f(y) &\geq f(\alpha x + (1-\alpha)y) + \alpha(1-\alpha)(0)f'(\alpha x + (1-\alpha)y)\end{aligned}$$

Which gives us,

$$\alpha f(x) + (1 - \alpha)f(y) \geq f(\alpha x + (1 - \alpha)y)$$

(3) $\Rightarrow$ (2) This is trivially true if we take $\alpha = \frac{1}{2}$

(2) $\Rightarrow$ (1). First we have (2) which is,

$$\frac{1}{2}f(x) + \frac{1}{2}f(y) \geq f\left(\frac{x+y}{2}\right)$$

Now take $x = x + h$ and $y = x - h$ which gives us,

$$\begin{aligned} f(x) &\leq \frac{1}{2}f(x+h) + \frac{1}{2}f(x-h) \\ 2f(x) &\leq f(x+h) + f(x-h) \\ f(x) - f(x-h) &\leq f(x+h) - f(x) \end{aligned}$$

Now note this holds true for any x, h and hence on the left side we can take y such that $y + h = x$ and $y = x - h$ such that $y \geq x$ when $h > 0$. This gives us,

$$\begin{aligned} f(y+h) - f(y) &\leq f(x+h) - f(x) \\ \frac{f(y+h) - f(y)}{h} &\leq \frac{f(x+h) - f(x)}{h} \end{aligned}$$

Taking limit of both sides when $h \rightarrow 0^+$ we have,

$$\begin{aligned} \lim_{h \downarrow 0} \frac{f(y+h) - f(y)}{h} &\leq \lim_{h \downarrow 0} \frac{f(x+h) - f(x)}{h} \\ f'(y) &\leq f'(x) \end{aligned}$$

for $y \leq x$. Now we are given that f'' is continuous and we just showed that f' is an increasing function. But this means that f'' is positive as mvt would tell us that,

$$\frac{f'(x) - f'(y)}{h} = f''(c)$$

And as $x \geq y$ which implies $f'(x) \geq f'(y)$ (as f' is increasing) we have $f'' \geq 0$. $\square$

Now lastly we need to show that,

$$f(x) = \sup\{g(x) : g \leq f \text{ on } \mathbb{R}, g \text{ is a linear polynomial}\}$$

It is enough to show that for each x_0 there is a linear polynomial $g(x)$ such that $f(x_0) = g(x_0)$ and $g \leq f$ on $\mathbb{R}$. First for a given point a consider the linear approximation of f at a . By definition the derivative tells us that,

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

Now this tell us that when x is arbitrarily close to a we have the approximation,

$$f(x) \approx f'(a)(x - a) + f(a)$$

Now define $g(x) = f'(a)(x - a) + f(a)$, here g is a linear approximation of f at a such that $g(a) = f'(a)(0) + f(a) = f(a)$. Now it is enough for us to show that $g \leq f$. Now, let x_0 be some arbitrary value. We need to show that for all x_0 we have $g(x_0) \leq f(x_0)$. Firstly, our approximation g at x_0 is equal to,

$$g(x_0) = f'(a)(x_0 - a) + f(a)$$

Now note that MVT gives us,

$$\frac{f(x_0) - f(a)}{x_0 - a} = f'(c) \quad \text{where } c \text{ is between } a \text{ and } x_0$$

Now consider two cases, first $a \leq x_0$. In this case we have $a \leq c \leq x_0$ which implies that $f'(a) \leq f'(c)$ as f' is an increasing function because f'' is positive by (1). Hence we can write,

$$\frac{f(x_0) - f(a)}{x_0 - a} = f'(c) \geq f'(a)$$

In the second case we have $x_0 \leq a$ and here we get $x_0 \leq c \leq a$ which means, $f'(c) \leq f'(a)$. Putting this we get,

$$\frac{f(a) - f(x_0)}{a - x_0} = f'(c) \leq f'(a)$$

In the first case we have,

$$\begin{aligned} g(x_0) &= f'(a)(x_0 - a) + f(a) \\ &\leq \frac{f(x_0) - f(a)}{x_0 - a}(x_0 - a) + f(a) \\ &= f(x_0) - f(a) + f(a) \\ g(x_0) &\leq f(x_0) \end{aligned}$$

And in the second case we have,

$$\begin{aligned} g(x_0) &= f(a) - f'(a)(a - x_0) \\ f'(a)(a - x_0) &= f(a) - g(x_0) \\ -g(x_0) &\geq -f(x_0) \\ g(x_0) &\leq f(x_0) \end{aligned}$$

So in both cases we show that for any x_0 we have $g(x_0) \leq f(x_0)$ which means that g is a linear function that approximates f at a and is smaller than equal to f in $\mathbb{R}$. As $g(a) = f(a)$ we have $g(a) \geq g'(a)$ where g' is any function smaller than f as if there were a g' such that $g'(a) > g(a)$ then we have $g'(a) > f(a)$ which implies that $g' \not\leq f$ at all points, a contradiction. So we have shown that,

$$f(x) = \sup\{g(x) : g \leq f \text{ on } \mathbb{R} \text{ where } g \text{ is a linear polynomial}\}$$